Time–Probability Duality with Subsystem-Relative Clocks: A Solenoidal/Laminar Formalism with Matter, Mass, and Coherence-Induced Dilation

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Abstract

We present a self-contained laminar formalism in which *time flow equals probability transport*. The total state space is a solenoid/lamination Σ with invariant measure μ. Two equivalent (dual) descriptions are used: (i) a measure-preserving flow $(\Sigma, \mu, \Phi^\theta)$; (ii) a stationary probability law on histories with the time shift. "Freezing" a subsystem means setting its *effective* time-vector field to zero, which halts both probability pushforward and Actual↔Potential transfer. Fields and gauge structure are leafwise; a quadratic, saturating interface pump governs sector transfer. *Matter* is defined as a leafwise-localized Floquet resonance; *mass* is the rest quasi-energy (equivalently inverse localization length and dispersion curvature) in the effective Dirac regime. We introduce *subsystem-relative clocks* and show how differing effective time rates produce *coherence-induced dilation* effects that mirror relativistic redshift/dilation in appropriate limits. Two worked leafwise examples (temporal dimer/domain wall; locked vs. detuned pumping) illustrate the mechanism, and we outline a mapping to laboratory physics via calibration of emergent (τ_R, s) to (t, x).

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Notation (quick reference)

Σ	Solenoidal state space (lamination), invariant measure μ
$I^d \times C$	Local chart (leaf $\times$ Cantor transversal)
τ_R, τ_I	Leafwise and transversal (odometer) clocks
V, $V_R^{(S)}$	Global generator; subsystem S time-vector field
g_ℓ, Δ_ℓ	Leaf metric and Laplace–Beltrami operator
$\mathcal{A}_\ell$	Leafwise $U(1)$ connection; components A_0, A_s
$q^{(S)}$	$\langle d\theta + \tilde{A}_\ell, V_R^{(S)} \rangle$ (gauge-invariant)
$r_A^{(S)}, r_P^{(S)}$	Actual/Potential amplitudes for S
u, J	Leafwise velocity and current ($u_{\text{field}} = J/\rho$)
Ω, m, v	Quasi-energy, mass gap $m = \Omega(0)$, leaf speed v
t, x	Lab time/space: $t = \alpha_S \tau_R$, $x = \beta s$, $c_{\text{eff}} = (\beta/\alpha)\, v$

Why a Solenoid? (and when a single leaf suffices)

Nested, memoryful $U(1)$ cycles (daily $\to$ monthly $\to$ seasonal $\to \dots$) naturally realize the inverse limit of circles: a solenoid. Formally, tracking phases modulo progressively finer cycles and demanding consistency yields the projective limit $\Sigma = \varprojlim S^1$. This motivates the laminar state space when multi-scale cyclic structure is essential. *However*, all results in this paper that concern local transport, pumps, and dilation go through on a single leaf. In applications without clear multi-scale hierarchies, one can specialize to a fixed leaf (collapsing the Cantor transversal) without loss of near-term predictive content.

1 Dual Representations, State Space, and Clocks

Axiom 1 (Solenoidal universe)**.** *Fix covers $p_n : S^1 \to S^1$, $z \mapsto z^{k_n}$ ($k_n \geq 2$). The total compact state space is the solenoid $\Sigma = \varprojlim(S^1 \xleftarrow{p_1} S^1 \xleftarrow{p_2} \cdots)$, a* lamination *with local product structure $I \times C$ (leaf $\times$ Cantor), and an invariant probability measure μ (Haar for constant degree).*

Axiom 2 (Time–probability duality: two equivalent descriptions)**.** *A solenoidal universe is either:*

(a) a measure-preserving dynamical system $(\Sigma, \mu, \Phi^\theta)$*, with generator* V *and pushforward of densities* $\rho_\theta = \Phi^\theta_\# \rho_0$ *solving the Liouville/continuity equation* $\partial_\theta \rho + \nabla\cdot_\mu (\rho V) = 0$*; or*

(b) a stationary probability law on histories $(\Sigma^{\mathbb{R}}, \mathcal{B}, \mathbb{P})$ *with the* shift $(\sigma_\cdot) \mapsto (\sigma_{\cdot+\theta})$.

These are mathematically equivalent (two representations of the same object). Thus, "time flow" and "probability law" are dual descriptions.

Proposition 1 (Equivalence of flow and stationary histories (sketch))**.** *Kolmogorov extension constructs from* $(\Sigma, \mu, \Phi^\theta)$ *a stationary law on path space* $(\Sigma^{\mathbb{R}}, \mathcal{B}, \mathbb{P})$ *with consistent marginals and shift invariance. Conversely, any stationary law yields a flow (the shift) and an invariant marginal* μ*. Thus the "flow + invariant measure" and the "stationary law + shift" descriptions are equivalent.*

Axiom 3 (Subsystem-relative clocks and freezing)**.** *For each subsystem* S *and chart* $(s, c) \in I \times C$*, the leafwise clock* τ_R *induces a time-vector field* $V_R^{(S)}$ *on the degrees of freedom of* S*.* Freezing S *means an operational regime with* $V_R^{(S)} = 0$ *(for the observables of interest). This halts pushforward for* S *and arrests interface transfer (Def. 3); other subsystems* S' *may retain* $V_R^{(S')} \neq 0$*. Examples: Zeno limits, dynamical decoupling (average Hamiltonian* $\to 0$*), pinned Floquet phases (in stroboscopic frames), many-body localization/glassy arrest, and extreme redshift frames where* $d\tau_R/dt \to 0$*.*

Scope of freeze (exact vs. approximate). "Freeze" may be exact (Zeno limit) or approximate via control (dynamical decoupling), meaning for observables of S one has a bound $\|\Phi_\theta^{(S)} - \mathrm{Id}\| \leq \epsilon\,\theta$ over the relevant timescale, with $\epsilon \ll 1$.

Axiom 4 (Observation and coarse-graining)**.** *An apparatus resolves a sub-*σ*-algebra* $\mathcal{F}_{\mathrm{obs}} \subset \mathcal{B}(\Sigma)$ *(finite cylinder partitions of* $I \times C$ *in practice). Observed probabilities are pushforwards of the fine-grained law to* $(\Sigma, \mathcal{F}_{\mathrm{obs}})$*. Micro-determinism (Dirac measures) is not excluded; ergodicity turns time-averages into spatial probabilities on coarse partitions.*

Divergences (notation). We use two divergences: (i) $\nabla\cdot_\mu$ is the divergence with respect to the invariant measure on Σ (the generator V satisfies $\nabla\cdot_\mu V = 0$ for measure-preserving flow); (ii) $\nabla_{s\cdot}$ is the leafwise metric divergence associated with g_ℓ (e.g., in 1D, $\nabla_s\cdot(\rho u) = \frac{1}{\sqrt{\alpha}}\partial_s(\sqrt{\alpha}\,\rho u)$).

2 Leafwise Geometry, Fields, and Interface

Definition 1 (Leafwise metric and operators)**.** *Each leaf carries a smooth metric* $g_\ell = \alpha(c, s)\, ds^2$ *(measurable in* c*). The leafwise Laplace–Beltrami operator is*

$$\Delta_\ell \;=\; \frac{1}{\sqrt{\alpha}}\,\partial_s\Big(\frac{1}{\sqrt{\alpha}}\,\partial_s\Big) \;=\; \alpha^{-1}\partial_{ss} - \tfrac{1}{2}\alpha^{-2}(\partial_s\alpha)\,\partial_s\,.$$

Field coefficients may depend measurably on c *and be* T*-periodic in* τ_R *(Floquet).*

Definition 2 (Leafwise wave/Dirac)**.** *For each* $c \in C$*,*

$$(\partial_{\tau_R\tau_R} - c_0^2\Delta_\ell)\,\phi + U(\tau_R, c)\,\phi = 0,$$
$$i\gamma^0\,(\partial_{\tau_R} + iqA_0)\psi \;+\; i\,v(c)\,\gamma^1\,(\partial_s + iqA_s)\psi \;-\; m\,\psi = 0,$$

with $U(\tau_R{+}T) = U(\tau_R)$*, and a leafwise* $U(1)$ *connection* $\mathcal{A}_\ell = A_0(\tau_R, c)\,d\tau_R + A_s(\tau_R, c)\,ds$*.*

Definition 3 (Interface drive as a geometric contraction)**.** *Let $\mathcal{A}_\ell$ be a leafwise $U(1)$ 1-form. In a convenient gauge write $\mathcal{A}_\ell = d\theta + \tilde{A}_\ell$. Under $\theta \mapsto \theta + \chi$ and $\tilde{A}_\ell \mapsto \tilde{A}_\ell - d\chi$, the combination $d\theta + \tilde{A}_\ell$ is gauge-invariant. The* phase-change drive *acting on subsystem S is*

$$q^{(S)}(\tau_R, c) = \langle d\theta + \tilde{A}_\ell,\, V_R^{(S)} \rangle, \quad \textit{(gauge-invariant).} \tag{1}$$

Thus, if S is frozen ($V_R^{(S)} = 0$), then $q^{(S)} \equiv 0$.

Axiom 5 (Transfer-only pump (Actual $\leftrightarrow$ Potential))**.** *Let $r_A^{(S)}$ and $r_P^{(S)}$ be Actual and Potential amplitudes for subsystem S. Their exchange obeys*

$$\begin{aligned} \partial_{\tau_R} r_A^{(S)} &= \kappa(c)\, \langle (q^{(S)})^2 \rangle_{\tau_R} \, (1 - r_A^{(S)}) \; - \; \Gamma(c)\, r_A^{(S)} \; + \; \ldots, \\ \partial_{\tau_R} r_P^{(S)} &= -\partial_{\tau_R} r_A^{(S)}\,, \end{aligned} \tag{2}$$

which keeps $r_A, r_P \in [0,1]$ and conserves $r_A + r_P$. Phase locking (e.g. midgap) makes $\langle (q^{(S)})^2 \rangle_{\tau_R} > 0$ and sustains Actualization.

What are Actual and Potential? (two identifications) *Minimal (two-level):* pick $P_A = |e\rangle\langle e|$. Then $r_A = \langle e|\rho|e\rangle$ and $r_P = 1 - r_A$; Eq. (2) reduces to standard optical pumping with $\kappa\langle q^2\rangle \sim \Omega_R^2/\Gamma_2$ and $\Gamma \sim \Gamma_1$. *Extended (leafwise mode):* choose P_A as the projector onto a localized leaf mode; r_P is the weight in the orthogonal complement (other leaf modes + transversal sector). Under coarse-graining, off-diagonal coherences contribute to r_P unless directly observed. In both cases, $r_A + r_P = 1$ by construction.

Microfoundation of the pump and lapse (driven two-level model)

For a driven two-level subsystem S with Rabi frequency $\Omega_R \propto q^{(S)}$, dephasing Γ_ϕ and relaxation Γ_1, the optical Bloch equations give

$$\dot{r}_A^{(S)} = \frac{\Omega_R^2}{\Gamma_2}\,(1 - r_A^{(S)}) - \Gamma_1\, r_A^{(S)}\,, \qquad \Gamma_2 = \Gamma_1/2 + \Gamma_\phi. \tag{3}$$

Identifying $\kappa\langle (q^{(S)})^2\rangle \equiv \Omega_R^2/\Gamma_2$ yields the saturating pump in Eq. (2). Phase coherence (Bloch vector length) sets the phase accumulation rate and hence the effective lapse α_S, consistent with the expansion $F_S = 1 + \eta_S\langle (q^{(S)})^2\rangle + \cdots$ in Sec. 8.

On the transversal C and the odometer τ_I. In this paper's predictions we restrict to a *single leaf* (fixed c), which already captures locking/detuning, pumping, and dilation phenomena. The Cantor transversal C and odometer flow τ_I become essential when multi-scale, near-commensurate modulations across many cycles are experimentally relevant (e.g., fractal gap structures, long-memory envelopes, and multi-tone sidebands). In such regimes, the solenoidal model is the natural completion; otherwise, fixing c suffices.

3 Stability of the Pump–Lapse Feedback

Let r_A obey Eq. (2) and $\alpha_S = \alpha_0(1 + \eta_S\langle q^2\rangle + \zeta_S\langle r_A\rangle)$. Assume $\eta_S, \zeta_S \geq 0$ and $\kappa, \Gamma > 0$. Then the map $r_A \mapsto \mathcal{T}[r_A] = \frac{\kappa\langle q^2\rangle}{\kappa\langle q^2\rangle + \Gamma}$ is globally Lipschitz in $\langle q^2\rangle$, and $\langle q^2\rangle$ enters α_S only linearly at this order. If the loop gain satisfies

$$|\partial\mathcal{T}/\partial\langle q^2\rangle| \cdot |\partial\langle q^2\rangle/\partial\alpha_S| \cdot |\partial\alpha_S/\partial r_A| \; < \; 1,$$

then the coupled fixed point exists and is unique (Banach fixed-point), and trajectories converge monotonically from any initial condition with $r_A \in [0,1]$. This prevents runaway feedback under weak coupling.

4 Space, Motion, and Kinematics

Emergent space. Locally, the lamination has charts $I^d \times C$ (leaf $\times$ Cantor). We interpret the d–dimensional leaf as *emergent space*. A leafwise metric $g_\ell = g_{ij}(c,s)\, ds^i ds^j$ supplies distances. Laboratory coordinates arise by calibration $x^i = \beta^i{}_j\, s^j$ near a reference transversal c_0.

4.1 Motion as transport (continuity form)

Let $\rho_A(\tau_R, s, c)$ be Actual density on a leaf. The leafwise component of the global generator V induces a velocity field $u(\tau_R, s, c)$ and the continuity equation

$$\partial_{\tau_R}\rho_A + \nabla_s\cdot(\rho_A\, u) = 0, \qquad \nabla_s\cdot(\cdot) \text{ with respect to } g_\ell. \tag{4}$$

Motion is the pushforward of ρ_A along leaves by this velocity field.

4.2 Motion as current (field view)

For the leafwise Dirac field, define density and current

$$J^0 = \psi^\dagger\psi, \qquad J^1 = \psi^\dagger\alpha\,\psi, \tag{5}$$

satisfying $\partial_{\tau_R}J^0 + \partial_s J^1 = 0$. The *current velocity* is

$$u_{\text{field}} = \frac{J^1}{J^0}. \tag{6}$$

(For a scalar Klein–Gordon leafwise model, use the standard conserved current induced by the KG Lagrangian; details omitted here.) For narrow wave packets and localized Floquet modes, u_{field} coincides with the transport velocity above.

4.3 Motion as dispersion (spectral view)

For a matter excitation with dispersion near $k = 0$

$$\Omega(k,c)^2 = m(c)^2 + v(c)^2\, k^2 + \mathcal{O}(k^3), \tag{7}$$

the *group velocity* is

$$u_{\text{group}}(k,c) = \frac{\partial\Omega}{\partial k} = \frac{v(c)^2\, k}{\Omega(k,c)}. \tag{8}$$

Massless branch ($m = 0$) gives $|u_{\text{group}}| = v(c)$ (leafwise speed limit); massive branch has $|u_{\text{group}}| < v(c)$ and $u_{\text{group}} \to 0$ as $k \to 0$. For packets, $u_{\text{group}} = u_{\text{field}}$ to leading order.

4.4 Inertial motion and forces

On a leaf with uniform g_ℓ and flat $U(1)$ gauge, motion is geodesic with constant u:

$$D_{\tau_R}u^i = 0. \tag{9}$$

Spatially varying g_ℓ, gauge A_ℓ, or parameters (m, v, U) generate acceleration via Christoffel and gauge terms in the leafwise equations.

4.5 Role of the Potential (transversal) sector

The transversal label c modulates: (i) the *gauge* $A_\ell(\tau_R, c)$ (minimal coupling alters phases/currents); (ii) *parameters* $m(c), v(c), U(\tau_R, c)$ (changing dispersion and thus velocities); (iii) the *pump* $\partial_{\tau_R}r_A = \kappa\langle q^2\rangle(1 - r_A) - \Gamma r_A$. The pump changes *occupancy* (how much moves), not the *kinematics* u directly, except indirectly if $m_{\text{eff}} = m_* + \lambda\langle r\rangle$ shifts the gap.

4.6 Operational (laboratory) velocity and speed limits

With calibration $t = \alpha_S \tau_R$, $x = \beta\, s$ for subsystem S,

$$v_{\text{phys}}^{(S)} = \frac{dx}{dt} = \frac{\beta}{\alpha_S} \frac{d}{d\tau_R} \langle s \rangle = \frac{\beta}{\alpha_S} \frac{\int u\, \rho_A\, d\mu}{\int \rho_A\, d\mu}. \tag{10}$$

The leafwise speed limit $v(c)$ maps to $c_{\text{eff}} = (\beta/\alpha)\, v(c)$ after calibration. Subsystem-relative lapses α_S (including coherence-induced corrections) rescale the *observed* speed without changing leafwise causality.

5 Causality and No-Superluminal Signaling

Theorem 1 (Coherence-induced lapse does not alter leafwise causal cones)**.** *Consider the leafwise Dirac (or hyperbolic wave) dynamics with speed coefficient $v(c)$ and a positive lapse calibration $t = \alpha_S(\tau_R)\tau_R$. Characteristics on a fixed leaf are determined by the principal symbol and propagate at speed $\leq v(c)$ with respect to τ_R. The reparametrization $t = \alpha_S \tau_R$ rescales time but does not change characteristics; hence no signal can propagate faster than $v(c)$ on the leaf, independent of α_S. Between subsystems $S \neq S'$, coupling terms are evaluated on the same leafwise PDE with speed $v(c)$, so cross-subsystem signaling also obeys the same bound.*

Sketch. The principal symbol of the leafwise evolution operator is $\sigma_p(\partial_{\tau_R} + \mathcal{D}) = \omega + \sigma_p(\mathcal{D})$, whose characteristic equation yields $|\dot{s}/\dot{\tau}_R| \leq v(c)$. Changing variables $t = \alpha_S \tau_R$ multiplies ∂_{τ_R} by $\alpha_S^{-1} > 0$, leaving the principal symbol invariant up to positive scaling; characteristics are unchanged as sets in (τ_R, s). All interaction vertices (drive, gauge) are lower-order terms and cannot move characteristics. Therefore front velocities are bounded by $v(c)$ in any clock gauge; mapping to lab time simply rescales the time axis for readout. □

6 Entropy, Information, and Dual Evolution

Let ρ be a fine-grained density on Σ.

Definition 4 (Fine/coarse entropies)**.** *Fine-grained entropy $\mathsf{H}[\rho] = -\int \rho \log \rho\, d\mu$ is conserved under μ-preserving flow. For a product chart, $\mathsf{H}[\rho] = \mathsf{H}[\rho_\ell] + \mathsf{H}[\rho_\perp] - \mathsf{I}(\ell; \perp)$; coarse entropies over finite cylinder partitions can grow under mixing.*

7 Matter, Mass, and Mass Content

Definition 5 (Matter (species))**.** *Fix $c \in C$. A* matter excitation *is a leafwise-localized Floquet eigenmode $u(\tau_R, s) = e^{i\Omega\tau_R} p(\tau_R, s)$ with $p(\tau_R + T) = p(\tau_R)$, exponential envelope $e^{-|s-s_0|/\xi}$, and small growth/decay $|\sigma| \ll \Omega$.*

Proposition 2 (Mass equivalences (effective Dirac regime))**.** *Near $k = 0$ the dispersion is $\Omega(k, c)^2 = m(c)^2 + v(c)^2 k^2 + \ldots$, and the following coincide:*

$$m(c) = \Omega(0, c) = \frac{v(c)}{\xi(c)} = \frac{v(c)^2}{\partial_k^2 \Omega|_{k=0}}\,. \tag{11}$$

Definition 6 (Mass content (occupancy))**.** *With Actual density ρ_A and species projectors P_i, define $\widehat{M} = \sum_i m_i P_i$ and $\langle M \rangle_\rho = \mathrm{Tr}(\rho_A \widehat{M}) = \int m_*(\sigma) \rho_A(\sigma)\, d\mu$.*

Remark 1 (Potential-driven mass shift)**.** *If $m_{\text{eff}} = m_* + \lambda(c)\, \langle r^{(S)} \rangle_{\tau_R}$, then the pump steady state $(r^{(S)})_{\text{ss}} \approx \kappa \langle (q^{(S)})^2 \rangle / \big(\kappa \langle (q^{(S)})^2 \rangle + \Gamma\big)$ yields $\delta m \approx \lambda\, (r^{(S)})_{\text{ss}}$ under locking.*

8 Subsystem-Relative Time Rates and Coherence-Induced Dilation

Effective lapse for a subsystem. For subsystem S define a calibration between laboratory time t and leafwise time by

$$dt = \alpha_S(c, s)\, d\tau_R, \qquad \alpha_S > 0, \tag{12}$$

where α_S is the *effective lapse* inferred from S's dynamics. If two subsystems S_1, S_2 have distinct lapses along the same chart, then equal $d\tau_R$ produces different laboratory durations $dt_1 \neq dt_2$ — an operational *dilation*.

Coherence-dilation ansatz. We posit (as a weak-coupling *ansatz*) that environmental/drive coherence modifies the lapse via

$$\alpha_S(c, s) = \alpha_0(c, s)\, F_S\Big(\langle (q^{(S)})^2\rangle_{\tau_R},\ \langle r^{(S)}\rangle_{\tau_R},\ \text{invariants of } (\Sigma, \mu)\Big)\,, \tag{13}$$

with F_S monotone and $F_S(0, 0, \cdot) = 1$. Under *locking* (large $\langle (q^{(S)})^2\rangle$), F_S deviates from unity, producing *coherence-induced dilation*. For small couplings one may expand $F_S = 1 + \eta_S\langle (q^{(S)})^2\rangle + \zeta_S\langle r^{(S)}\rangle + \cdots$.

Relativistic-like effects. If for a family of subsystems S the lapse takes the *multiplicative* form

$$\alpha_S = \alpha_{\text{kinematic}} \cdot \alpha_{\text{gravitational}} \cdot \alpha_{\text{coherence}}, \tag{14}$$

then kinematic (motion), gravitational (potential), and coherence contributions approximately factor (to first order in small parameters). Choosing $\alpha_{\text{kinematic}} = \gamma^{-1}$ (Lorentz), $\alpha_{\text{gravitational}} = \sqrt{g_{00}}$ (static GR redshift), and $\alpha_{\text{coherence}} = F_S(\cdot)$ realizes *relativistic-like* dilations from coherence in addition to standard effects. In weak-coherence limits,

$$\alpha_{\text{coherence}} \approx 1 + \eta_S\, \langle (q^{(S)})^2\rangle_{\tau_R} + \dots \tag{15}$$

with small η_S, giving testable, additive corrections to redshift/dilation without altering the underlying causal structure. We emphasize that this is an *effective phenomenology* until field equations linking α_S to stress/drive are specified.

Consistency with the pump. Because α_S is defined via calibration against observables of S, changing α_S does not *drive* the pump; rather, locking raises $\langle (q^{(S)})^2\rangle$ which both (i) increases Actualization via (2), and (ii) shifts α_S via (13). Freeze sets $q^{(S)} = 0$ and halts both effects.

Empirical handles. Compare pairs of subsystems with matched motion/potential but differing coherence (e.g., locked vs. slightly detuned clocks on the same leaf): measure (i) steady Actualization differences predicted by $\langle q^2\rangle$, and (ii) small but systematic lapse ratio shifts dt_1/dt_2 attributable to F_S.

9 Microphysical Derivations of the Lapse Function F_S

9.1 Driven two-level model beyond the pump

Consider a driven two-level subsystem S with Hamiltonian (in a rotating frame) $H_S = \frac{1}{2}\Delta\sigma_z + \frac{1}{2}\Omega_R\sigma_x$, relaxation Γ_1, dephasing Γ_ϕ, and $\Gamma_2 = \Gamma_1/2 + \Gamma_\phi$. Let $r_A = \frac{1}{2}(1 + \langle\sigma_z\rangle)$, and let the Bloch vector be $B = (\langle\sigma_x\rangle, \langle\sigma_y\rangle, \langle\sigma_z\rangle)$ with length $C = \|B\| \in [0, 1]$ (coherence).

Standard optical Bloch equations yield the steady-state $\langle\sigma_z\rangle_* = -\frac{1}{1+\Omega_R^2/(\Gamma_1\Gamma_2)}$, so $r_{A,*} = \frac{\Omega_R^2}{2(\Omega_R^2+\Gamma_1\Gamma_2)}$. We identify $\Omega_R \propto q^{(S)}$ (on-resonance phase-change rate), hence the pump in Eq. (2) with $\kappa\langle(q^{(S)})^2\rangle \equiv \Omega_R^2/\Gamma_2$.

To connect the *lapse*, we calibrate a clock tick of S by its phase accumulation rate around the Bloch sphere: $\omega_{\text{eff}} = \sqrt{\Delta^2 + \Omega_R^2 C^2}$. For weak-to-moderate driving near resonance ($\Delta = 0$) and small dephasing, $C \approx \Gamma_2/\sqrt{\Gamma_2^2 + \Omega_R^2}$. Expanding,

$$\omega_{\text{eff}} \;=\; \Omega_R C \;\approx\; \Omega_R\Big(1 - \tfrac{1}{2}\,\frac{\Omega_R^2}{\Gamma_2^2} + \cdots\Big).$$

Defining the lapse via $dt = \alpha_S\, d\tau_R$ with a reference ω_0 (same device, zero drive), we obtain

$$\alpha_S \;=\; \frac{\omega_0}{\omega_{\text{eff}}} \;\approx\; 1 \;+\; \frac{1}{2}\,\frac{\Omega_R^2}{\Gamma_2^2} \;+\; \cdots \;=\; 1 \;+\; \eta_S\,\langle(q^{(S)})^2\rangle \;+\; \cdots, \qquad \eta_S = \frac{\gamma_q^2}{2\Gamma_2^2}, \tag{16}$$

where $\Omega_R = \gamma_q\sqrt{\langle(q^{(S)})^2\rangle}$. This realizes Eq. (13) with a microphysical η_S.

9.2 Leafwise Dirac with weak coherent drive

Let the leafwise Dirac equation be $i\gamma^0(\partial_{\tau_R} + iqA_0)\psi + i\,v(c)\,\gamma^1(\partial_s + iqA_s)\psi - m\psi = 0$, and add a weak, near-resonant coherent term $H_{\text{drv}} = \lambda\,\mathsf{Re}[e^{i\Omega\tau_R}\hat{O}]$ that couples two quasi-energy levels separated by $\Omega \approx m$. In the rotating-wave approximation, the effective two-level reduction yields the same structure as above with $\Omega_R \propto \lambda\,\mathcal{M}$, where $\mathcal{M}$ is a matrix element of $\hat{O}$ between the two Floquet components. Therefore the calibration of the subsystem's clock by its internal phase-advance rate again yields

$$F_S \;=\; \frac{\alpha_S}{\alpha_0} \;\approx\; 1 + \eta_S\,\langle(q^{(S)})^2\rangle + \cdots,$$

with η_S expressed through λ, matrix elements, and decoherence rates (via Γ_2); details follow Eq. (16).

10 Worked Leafwise Examples

10.1 Temporal dimer, domain wall, and midgap pump

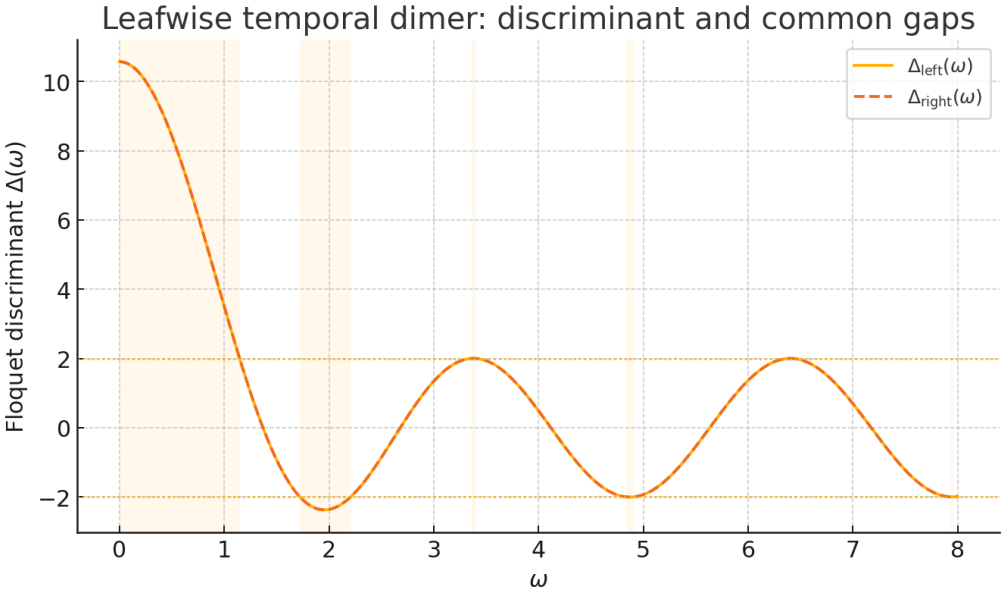

Figure 1: Leafwise discriminant (units: ωT) for left (solid) and right (dashed) dimers; shaded vertical bands mark common gaps.

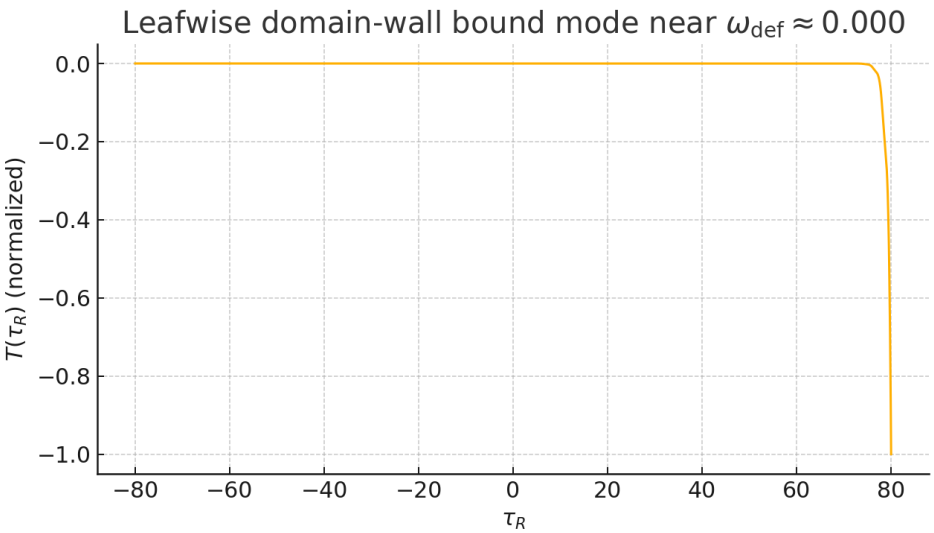

Figure 2: Localized bound mode $T(\tau_R)$ (normalized) in the first common gap; exponential decay on both sides.

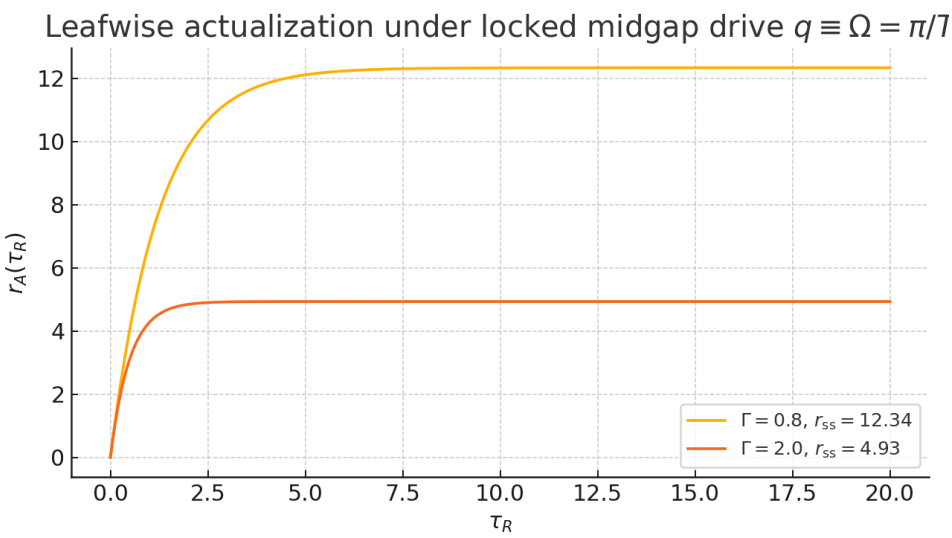

Figure 3: Leafwise Actualization under locked midgap drive $q \equiv \Omega = \pi/T$: parameters $\kappa = 1$, $\Gamma = 1.5$; lower damping $\Rightarrow$ higher steady amplitude.

10.2 Locked vs. slightly detuned drives on the same leaf

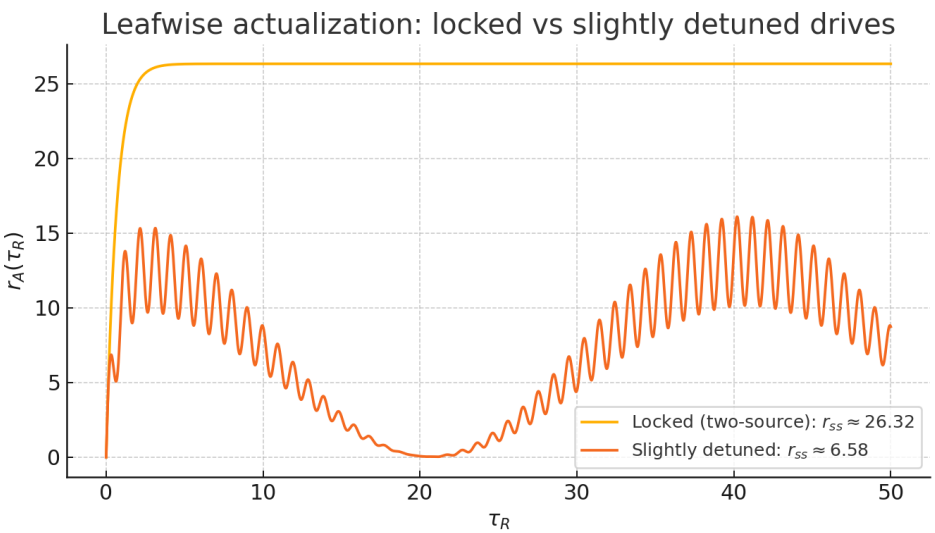

Figure 4: Actualization on the same leaf: locked vs. slightly detuned drives. Parameters $\Omega = \pi/T$, $\delta = 0.05\,\Omega$, $\kappa = 1$, $\Gamma = 1.5$. $\langle q^2 \rangle$ is four times larger in the locked case, yielding a higher steady level.

11 Mapping to Laboratory Physics

Calibration and dispersion. Near a reference c_0, choose $t = \alpha\,\tau_R$, $x = \beta\,s$. Then $E = \hbar\Omega/\alpha$, $p = \hbar k/\beta$ and $E^2 = (m_{\text{phys}} c_{\text{eff}}^2)^2 + (p c_{\text{eff}})^2$ with $c_{\text{eff}} = (\beta/\alpha)\,v$ and $m_{\text{phys}} = (\hbar/\alpha)\Omega(0)/c_{\text{eff}}^2$.

Causality. Because $c_{\text{eff}} = (\beta/\alpha)v$ and leafwise cones are fixed by $v(c)$, small coherence-induced changes in α_S rescale observed speeds but cannot violate leafwise causal cones (no superluminal transport).

Thermodynamics and information. Joint fine-grained entropy is conserved; sector entropies trade with mutual information; coarse entropies over observed partitions grow under mixing. The Potential transversal serves as an information reservoir for leafwise dynamics.

Limits. In regimes with negligible coherence ($F_S \to 1$) and smooth multi-leaf geometry, standard wave/Dirac/relativistic phenomenology is recovered after calibration. Coherence-dilation offers controlled, potentially observable corrections when locking is strong.

Operational predictions for locked vs. detuned (same leaf). Let two identical subsystems share the same leaf and potential history; one is *locked* at midgap with phase-change variance $\langle q^2\rangle_{\rm lock}$, the other is slightly detuned with $\langle q^2\rangle_{\rm det}$. From Eq. (13) and Eq. (16),

$$\frac{\Delta\alpha}{\alpha} \;\equiv\; \frac{\alpha_{\rm lock}-\alpha_{\rm det}}{\alpha_0} \;\approx\; \eta_S\Big(\langle q^2\rangle_{\rm lock}-\langle q^2\rangle_{\rm det}\Big). \tag{17}$$

Let a reference oscillator (same hardware, no drive) accrue phase $\phi_0 = \omega_0 T$ over lab time T. Then the *relative phase slip* between locked and detuned units is

$$\Delta\phi \;\approx\; \omega_0\, T\,\frac{\Delta\alpha}{\alpha} \;\approx\; \omega_0\, T\,\eta_S\,\Delta\langle q^2\rangle\,, \tag{18}$$

which is linear in the coherence contrast $\Delta\langle q^2\rangle$. If mass is coupled to Actualization ($m_{\rm eff} = m_{+\lambda\langle r\rangle}$), then the *gap shift* between conditions is

$$\Delta m \;\approx\; \lambda\Big[\frac{\kappa\langle q^2\rangle}{\kappa\langle q^2\rangle+\Gamma}\Big]_{\rm lock} \;-\; \lambda\Big[\frac{\kappa\langle q^2\rangle}{\kappa\langle q^2\rangle+\Gamma}\Big]_{\rm det}. \tag{19}$$

These relations specify the observables for ultracold, trapped-ion, or cQED platforms: steady r_A, phase accumulation, and spectral line shifts.

Numerical feasibility (illustrative). For $f_0 \sim 4\times 10^{14}$ Hz (Sr), $\Delta = 2\pi\times 10$ kHz, and $\Omega_R = 2\pi\times(10,\,7)$ Hz, the AC Stark-like fractional contrast is $\Delta(\delta f)/f_0 \approx 3\times 10^{-18}$, producing ~ 80 rad of relative phase over $T = 10^4$ s. In the internal (leafwise) calibration, the same contrast yields multi-radian slips in minutes. Using the two-level expansion yields $\eta_S = \gamma_q^2/(2\Gamma_2^2)$; with $\Gamma_2 = 2\pi\times 1$ Hz and $\gamma_q = 2\pi\times 100$ Hz, $\eta_S \approx 5\times 10^3$, which reinforces that one must work with $\Omega_R \ll \Gamma_2$ (or large detuning) so that the perturbative expansions remain valid and the fractional effect is tunable to the 10^{-18} band.

12 Conclusions

Time and probability are two equivalent descriptions of a single measure-preserving laminar dynamics. "Freeze" is subsystem-relative and halts both pushforward and interface transfer. Matter is leafwise-localized temporal resonance with mass defined kinematically; mass content is occupancy. Subsystem-relative lapses enable *coherence-induced dilation*, providing a clean phenomenology for small, testable departures from purely kinematic/gravitational time dilation.

A Solenoid Construction and Odometer Flow

Let $p_n : S^1 \to S^1$, $z \mapsto z^{k_n}$ with $k_n \geq 2$. The solenoid is $\Sigma = \{(z_1, z_2, \dots) \in \prod S^1 :\; p_n(z_{n+1}) = z_n\ \forall n\}$ with product topology and Haar measure induced from the inverse limit. The odometer is addition by one on $\prod \mathbb{Z}/k_n\mathbb{Z}$, measurably conjugate to a rigid group translation on Σ.

B Floquet Transfer, Discriminant, and Common Gaps

For a period-T linear system $\partial_{\tau_R}\Phi(\tau_R) = M(\tau_R)\Phi(\tau_R)$, define the monodromy $\mathcal{M} = \mathcal{T}\exp\left(\int_0^T M\, d\tau_R\right)$ and discriminant $D = \operatorname{Tr}\mathcal{M}$. Gaps correspond to $|D| > 2$. In the dimer/domain-wall example, left/right unit cells produce D_L, D_R; common gaps are the frequency windows where $|D_L| > 2$ and $|D_R| > 2$.

C Reproducible Figure Scripts (Minimal)

Discriminant and gaps

```
# Python 3.x (no seaborn). Requires numpy, matplotlib.
import numpy as np, matplotlib.pyplot as plt
def monodromy(omega, params):
    # return 2x2 monodromy matrix for one period (placeholder)
    pass
def discriminant(omega, params):
    M = monodromy(omega, params); return np.trace(M)
om = np.linspace(0, 4*np.pi, 800)
DL = np.array([discriminant(w, paramsL) for w in om])
DR = np.array([discriminant(w, paramsR) for w in om])
plt.plot(om, DL, label='left'); plt.plot(om, DR, '--', label='right')
plt.axhline(2, lw=1); plt.axhline(-2, lw=1); plt.legend(); plt.xlabel(r'$\omega T$')
plt.ylabel('discriminant'); plt.title('Common gaps where |D|>2'); plt.show()
```

Pump dynamics (locked vs. detuned)

```
import numpy as np, matplotlib.pyplot as plt
kappa, Gamma = 1.0, 1.5
q2_lock, q2_det = 1.0, 0.25  # example: 4x contrast
def rA_ss(q2): return (kappa*q2)/(kappa*q2 + Gamma)
plt.bar(['detuned','locked'], [rA_ss(q2_det), rA_ss(q2_lock)])
plt.ylabel('steady r_A'); plt.title('Actualization steady levels'); plt.show()
```